ELEC5305 Project Proposal

1. Project Title

Deep Learning-Assisted Dynamic Binaural Music Spatialisation

2. Student Information

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3. Project Overview

This project aims to develop a music spatialisation system that transforms an ordinary stereo music recording into a dynamic binaural audio experience suitable for playback through standard stereo headphones. The main challenge is that instruments and vocals in a commercial music track are normally mixed together, which prevents them from being repositioned independently.

To address this problem, a pretrained deep-learning music source-separation model will first separate the input track into several stems, such as vocals, drums, bass and other instruments. Each stem will then be independently positioned in a virtual acoustic space using head-related transfer function (HRTF) and head-related impulse response (HRIR) processing. In addition to fixed positions, the project will implement dynamic source movement, allowing individual musical components to move smoothly around the listener over time.

The main focus is the implementation of the complete signal-processing pipeline rather than the development of a new deep-learning architecture. A simple user interface will allow users to load music, assign spatial positions or movement paths, generate the binaural mix and export the final stereo audio.

4. Background and Motivation

Modern music recordings already contain stereo positioning and artificial reverberation, but conventional stereo panning mainly controls the relative levels of the left and right channels. Human spatial hearing is more complex and uses binaural cues including interaural time differences (ITD), interaural level differences (ILD), and frequency-dependent filtering caused by the head and outer ears.

HRTFs describe how sound arriving from a particular direction is modified before reaching each ear. By convolving a source signal with corresponding left- and right-ear HRIRs, it is possible to encode these spatial cues into an ordinary two-channel audio signal. Such binaural signals can be reproduced using standard headphones without requiring specialised spatial-audio hardware.

A limitation is that an already mixed song cannot easily position individual instruments independently. Recent deep-learning source-separation systems such as Demucs can estimate vocals, drums, bass and other sources from a stereo mixture. Combining source separation with binaural signal processing therefore provides a practical way to spatially remix existing music.

This topic was selected because it integrates music, deep learning and core audio signal-processing concepts while producing results that can both be quantitatively analysed and directly heard.

5. Proposed Methodology

Python will be used as the main implementation platform because it provides suitable libraries for deep learning, audio processing and interface development. A pretrained music source-separation model, initially expected to be a Demucs-based model, will be used to estimate four stems:

$$\boldsymbol{x(t)} \rightarrow \hat{\boldsymbol{s}}_{\mathbf{vocal}}\boldsymbol{(t)}, \hat{\boldsymbol{s}}_{\mathbf{drums}}\boldsymbol{(t)}, \hat{\boldsymbol{s}}_{\mathbf{bass}}\boldsymbol{(t)}, \hat{\boldsymbol{s}}_{\mathbf{other}}\boldsymbol{(t)}$$

MUSDB18 will be used for source-separation testing and quantitative evaluation, while additional stereo music tracks may be used for listening demonstrations.

For binaural rendering, public HRTF/HRIR data such as the CIPIC HRTF Database will be investigated. For a source $\left(s_i(t)\right)$ placed at azimuth(θ_i), the basic binaural rendering operation is

$$\boldsymbol{y_{L,i}(t) = s_i(t) * h_L(t, \theta_i)}$$

$$\boldsymbol{y_{R,i}(t) = s_i(t) * h_R(t, \theta_i)}$$

where (h_L) and (h_R) are the HRIRs for the left and right ears. The processed stems will then be combined to produce the final binaural stereo signals.

The HRTF stage will not be treated as a black-box library call. HRIR data loading, coordinate mapping, convolution, sampling-rate handling, gain management and binaural mixing will be implemented and analysed as core technical components.

Dynamic spatialisation will extend the static case by allowing

$$\boldsymbol{\theta} = \boldsymbol{\theta}(\boldsymbol{t})$$

so that sources can move around the listener. Because HRTF databases contain measurements only at discrete positions, the music will be processed in frames or blocks and neighbouring HRIRs will be interpolated or crossfaded. This is intended to reduce clicks, abrupt spectral changes and perceptual jumps when a source moves between locations.

After spatial processing, stems will be synchronised and mixed with appropriate gain control, headroom and clipping prevention. A lightweight user interface will allow selection of an input track and specification of static or dynamic positions for each stem. The initial implementation will be offline rather than real-time to keep the project achievable within one semester.

Evaluation will represent approximately 20% of the project. Source-separation quality will be assessed using metrics such as SDR or SI-SDR where ground-truth stems are available. Spatial processing will be evaluated using waveform, frequency-spectrum and spectrogram comparisons, together with ILD and ITD measurements. Dynamic tests will examine whether the generated binaural cues vary consistently with the requested source trajectory. A small comparison with conventional stereo panning may also be included to illustrate the additional binaural cues introduced by HRTF processing.

6. Expected Outcomes

The expected outcome is a working prototype capable of:

- separating an ordinary music recording into multiple stems using deep learning;
- independently positioning each stem using HRTF-based binaural processing;
- producing both static and dynamically moving sound sources;
- combining the processed stems into a playable stereo binaural mix;
- exporting processed audio and providing a simple interface for controlling source positions.

The final report will include source-separation metrics, waveform and spectrogram results, frequency-domain analysis, ILD/ITD measurements, dynamic movement examples and listening demonstrations. Example original, separated and spatialised audio files will also be provided through the GitHub repository.

7. Timeline

Week	Planned Work
1-2	Define project topic and scope
3-5	Literature review; investigate **MUSDB18, Demucs and HRTF/HRIR datasets**
6	Implement and test pretrained music source separation
7	Analyse separated stems and establish baseline evaluation
8	Implement static **HRIR convolution** and binaural rendering
9	Implement multi-stem spatial mixing and audio export
10	**Develop dynamic positioning, block processing and HRTF interpolation/crossfading**
11	Develop simple UI and complete quantitative evaluation
12	Optimise processing, generate figures and audio demonstrations
13	Complete final report, GitHub documentation and project demonstration

8. References

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[3] Wightman, F. L., & Kistler, D. J. (1989). Headphone simulation of free-field listening. I: Stimulus synthesis. *The Journal of the Acoustical Society of America, 85*(2), 858–867.

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[5] Rafii, Z., Liutkus, A., Stöter, F.-R., Mimilakis, S. I., & Bittner, R. (2017). *The MUSDB18 Corpus for Music Separation*. Zenodo.